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**Computer System for Processing and Analysis of Eye Data to Support  
Laser Coagulation in the Treatment of Diabetic Retinopathy**

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## **NORMATIVE REFERENCES**

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## DESIGNATIONS AND ABBREVIATIONS

|          |                                                                                                                                                       |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI       | Artificial Intelligence                                                                                                                               |
| ALO      | Attention–Lesion Overlap (explainability metric)                                                                                                      |
| APTOS    | Asia Pacific Tele-Ophthalmology Society (2019 Blindness Detection dataset)                                                                            |
| AUC      | Area Under the Curve                                                                                                                                  |
| BYOL     | Bootstrap Your Own Latent (self-supervised learning method)                                                                                           |
| CAM      | Class Activation Mapping                                                                                                                              |
| CFC-n    | Claim Formulation Constraint n — a rule governing the form a claim may take; the CFC-2 series enumerates claim types this dissertation forbids itself |
| CLAHE    | Contrast-Limited Adaptive Histogram Equalization                                                                                                      |
| CNN      | Convolutional Neural Network                                                                                                                          |
| CNR      | Contrast-to-Noise Ratio                                                                                                                               |
| DDR      | Diabetic Retinopathy grading dataset (DDR)                                                                                                            |
| DGL-n    | Deployment and Generalization Limitation n — a bound on how far a result may be carried beyond the conditions under which it was obtained             |
| DINO     | self-Distillation with No labels (self-supervised learning method)                                                                                    |
| DR       | Diabetic Retinopathy                                                                                                                                  |
| ECE      | Expected Calibration Error                                                                                                                            |
| EH-n     | Evidence Hierarchy rule n — the ranking of metrics by evidentiary weight, and the criteria a result must meet before it counts as decisive            |
| EHR      | Electronic Health Record                                                                                                                              |
| EMD      | Earth Mover's Distance                                                                                                                                |
| EyePACS  | Eye Picture Archive Communication System (diabetic-retinopathy dataset)                                                                               |
| FOV      | Field of View                                                                                                                                         |
| GDPR     | General Data Protection Regulation                                                                                                                    |
| Grad-CAM | Gradient-weighted Class Activation Mapping                                                                                                            |
| HIPAA    | Health Insurance Portability and Accountability Act                                                                                                   |
| IDRiD    | Indian Diabetic Retinopathy Image Dataset                                                                                                             |
| IoU      | Intersection over Union                                                                                                                               |
| KL       | Kullback–Leibler divergence                                                                                                                           |
| LAB      | CIELAB colour space                                                                                                                                   |
| MLP      | Multilayer Perceptron                                                                                                                                 |
| MoCo     | Momentum Contrast (self-supervised learning method)                                                                                                   |
| NC-n     | Non-Claim n — a statement this dissertation explicitly does not make, recorded so that its results are not read as making it                          |
| NPV      | Negative Predictive Value                                                                                                                             |
| OD       | Optic Disc                                                                                                                                            |
| OD-n     | Operational Definition n — the definition by which a term of this dissertation is measured; distinct from OD above, which is never                    |

|             |                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------|
|             | written with a number                                                                                                   |
| ODIR-5K     | Ocular Disease Intelligent Recognition dataset (5,000 patients)                                                         |
| PACS        | Picture Archiving and Communication System                                                                              |
| PC-n        | Primary Claim n — a numbered claim of the dissertation's argument, each carrying its evidence and its assessed strength |
| PPV         | Positive Predictive Value                                                                                               |
| ReLU        | Rectified Linear Unit                                                                                                   |
| RFMiD       | Retinal Fundus Multi-disease Image Dataset                                                                              |
| RIQA        | Retinal Image Quality Assessment                                                                                        |
| ROC         | Receiver Operating Characteristic                                                                                       |
| SB-n        | Scope Boundary n — a statement of what lies outside what this dissertation claims                                       |
| SimCLR      | Simple framework for Contrastive Learning of Representations                                                            |
| SIR-n       | Source Interpretation Rule n — a rule constraining what may be attributed to a cited source                             |
| SSIM        | Structural Similarity Index                                                                                             |
| SSL         | Self-Supervised Learning                                                                                                |
| STARE       | Structured Analysis of the Retina (dataset)                                                                             |
| UML         | Unified Modeling Language                                                                                               |
| VRAM        | Video Random-Access Memory                                                                                              |
| VVI         | Vessel Visibility Index                                                                                                 |
| WSL         | Windows Subsystem for Linux                                                                                             |
| XAI         | Explainable Artificial Intelligence                                                                                     |
| D           | Field-of-view diameter, in pixels                                                                                       |
| CL          | Clip limit of the CLAHE transform                                                                                       |
| F1          | F1-score (harmonic mean of precision and recall)                                                                        |
| G           | Generalization ratio, $G = F1_{\text{external}} / F1_{\text{EyePACS}}$                                                  |
| T           | Global threshold of the dual-constraint CLAHE (T/80 formulation)                                                        |
| $\alpha$    | Class-weight parameter of the Focal Loss (inverse-frequency)                                                            |
| $\gamma$    | Focusing parameter of the Focal Loss ( $\gamma = 2$ )                                                                   |
| $\kappa$    | Cohen's Kappa with quadratic weights                                                                                    |
| $\sigma$    | Standard deviation of the Gaussian kernel in flat-field correction, $\sigma = 0.07 \cdot D$                             |
| $\Delta F1$ | Cross-dataset F1 drop, $\Delta F1 = F1_{\text{EyePACS}} - F1_{\text{external}}$                                         |

## DEFINITIONS

**Architectural complexity** — the capacity of a CNN as defined by the number of convolutional layers, total trainable-parameter count, filter-size range, and the presence or absence of regularization components (batch normalization, dropout).

**Baseline (baseline arm)** — the reference configuration of Experiment 1: stretch-resize to  $512 \times 512$  followed by ImageNet normalization, producing a 3-channel tensor with no field-of-view mask and no preprocessing stages; the backbone is initialized from ImageNet.

**CLAHE (Contrast-Limited Adaptive Histogram Equalization)** — an adaptive contrast-enhancement method operating on local image tiles with a clip limit that bounds contrast amplification; applied in the pipeline to the LAB L-channel under a dual-constraint clip limit (Stage 5).

**Composite independent variable** — the combined manipulated factor of Experiment 1 (H-1), in which the full-pipeline arm differs from baseline jointly along the preprocessing axis (eight-stage pipeline vs. stretch-resize + ImageNet normalize) and the pretraining axis (ophthalmology-specific self-supervised pretraining vs. ImageNet).

**Convolutional Neural Network (CNN)** — a neural architecture comprising convolutional and pooling layers for feature extraction and fully connected layers for classification; the classification backbone operating on preprocessed fundus images for five-class DR staging.

**Cross-validation** — a validation strategy using 5-fold patient-level stratified splitting; for each fold, four folds serve as training data and one as test data, with no patient's images appearing in both partitions; metrics are reported as mean  $\pm$  standard deviation across folds.

**Data augmentation** — image transformations (unified affine, ColorJitter [brightness/contrast/saturation/hue], Gaussian noise, JPEG compression) applied to the training data to increase variability and improve generalization; the train-only Stage 6 of the pipeline.

**Dataset-specific normalization** — channel-wise normalization (Stage 7) using mean and standard deviation computed from the training split of the primary dataset rather than ImageNet statistics; the normalization of the full pipeline.

**Diabetic retinopathy (DR)** — a microvascular complication of diabetes mellitus and a leading cause of preventable vision loss, graded in five severity stages (0 — none, 1 — mild, 2 — moderate, 3 — severe, 4 — proliferative).

**Diagnostic effectiveness** — the joint performance profile on four primary metrics (accuracy, weighted F1-score, ROC-AUC, and Cohen's Kappa with quadratic weights) computed on the held-out test partition.

**Domain shift** — the distributional difference between training data and target deployment data arising from differences in imaging equipment, patient populations, or acquisition protocols.

**Field-of-view (FOV) mask** — a binary spatial mask (1.0 = real fundus data, 0.0 = zero-padding) generated at Stage 3 and appended as the 4th input channel, allowing the CNN to distinguish genuine retinal pixels from padded background.

**Flat-field correction (adaptive)** — Gaussian-blur subtraction with adaptive  $\sigma = 0.07 \cdot D$  ( $D$  = FOV diameter), applied inside the FOV mask to normalize uneven illumination while preserving local contrast; Stage 4 of the pipeline.

**Fundus image** — a retinal photograph acquired by fundoscopy; the exclusive imaging modality of the dissertation and the input to the preprocessing pipeline and the CNN classifier.

**Generalization (cross-database)** — the ratio of test-set F1-score on a secondary dataset to test-set F1-score on the primary dataset under the same trained model without retraining; the generalization ratio  $G = F1\_external / F1\_EyePACS$ .

**Grad-CAM (Gradient-weighted Class Activation Mapping)** — an explainability method producing class-discriminative localization maps from a gradient-weighted combination of final convolutional feature maps; an interpretability tool, not a clinical localization of pathology.

**Image quality** — the measurable capacity of a fundus image to support automated detection of microvascular features relevant to DR staging, assessed through downstream classification performance rather than a subjective visual score.

**Integrated pipeline dominance (H-1)** — the primary hypothesis that the integrated full-pipeline configuration outperforms the baseline as a unitary system, validated when the empirical dominance criterion ( $\geq 5$  pp weighted-F1 gain,  $\geq 0.02$  ROC-AUC gain, no Cohen's Kappa degradation) is satisfied.

**Ophthalmology-specific self-supervised pretraining** — self-supervised pretraining of the CNN backbone on an unlabeled retinal fundus corpus (no DR labels), used as the full-pipeline-arm initialization; learns fundus-domain representations directly on the 4-channel pipeline tensor.

**Physician-in-the-loop** — a design paradigm in which the clinician acts not merely as a consumer of AI outputs but as a tuner and auditor who interprets results and adjusts the system; the proposed system is a decision-support tool within this paradigm.

**Preprocessing pipeline** — the canonical ordered eight-stage sequence (Stages 0–7) applied to fundus images prior to CNN input; the full pipeline applies all eight stages, producing a 4-channel tensor (RGB + FOV mask).

**Resource-limited environment** — a deployment context characterized by at least two of: absence of GPU acceleration; available RAM below 16 GB; batch-processing-time constraints; network connectivity precluding continuous cloud-API reliance.

**Transfer learning** — the reuse and adaptation of pretrained network weights to the fundus-image domain; under H-1 the initialization source is slaved to the arm (ImageNet for baseline, ophthalmology-specific SSL for the full pipeline).

**Weighted loss function** — a cross-entropy loss with class-specific weights addressing class imbalance and exploiting the ordinal structure of the five-class DR grading.